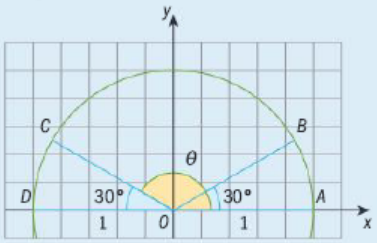
Investigation

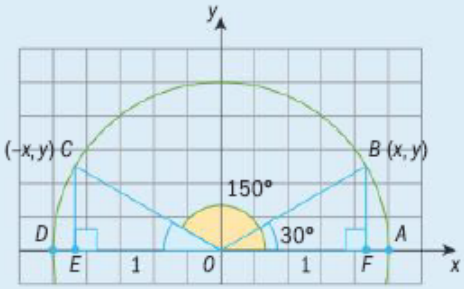
Investigation – obtuse angles

This diagram shows point B at a positive angle of 30° from OA, and point C at a positive angle of θ from OA.



Find the value of θ.
What are the coordinates of point B?
Use the symmetry of the unit circle to write down the coordinates of point C.

Now look at the triangles formed by the sides OB and OC and the x-axis.



$\triangle EOC$ is congruent to $\triangle FOB$. Both are 30°– 60°– 90° triangles with hypotenuse length 1. You can also see that if the coordinates of point B are (x, y), then the coordinates of point C are $(-x, y)$.

The coordinates of B are $(\cos 30°, \sin 30°)$, or $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$.

So the coordinates of point C are $(\cos 150°, \sin 150°)$, which are the same as the coordinates $(-\cos 30°, \sin 30°)$, or $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$.

Draw diagrams showing each of these pairs of angles in the unit circle.

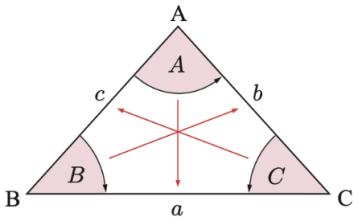
1 40° and 140°
2 25° and 155°
3 68° and 112°

Label the coordinates of the points where the non-horizontal sides meet the unit circle. What do you notice?

I- Area of triangles

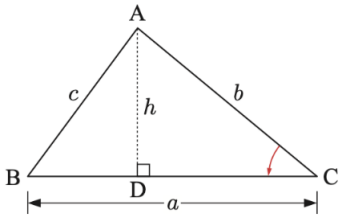
Property

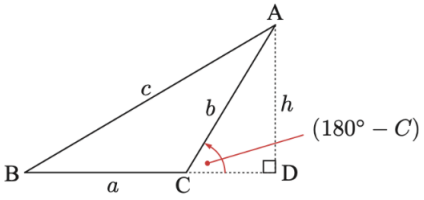
Triangle ABC has angles of size A, B, and C. The sides opposite these angles are labelled a, b, and c respectively.



$$\text{Area} = \frac{1}{2}ab\sin C.$$

Proof





In both triangles the altitude h is constructed from A to D on $[BC]$ (extended if necessary).

$$\sin C = \frac{h}{b} \qquad \therefore \qquad h = b\sin C$$

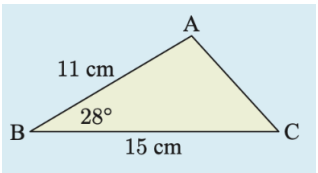
$$\sin(180^\circ - C) = \frac{h}{b} \qquad \therefore \qquad h = b\sin(180^\circ - C)$$

But

$$\sin(180^\circ - C) = \sin C \qquad \therefore \qquad h = b\sin C.$$

Example

Find the area of triangle ABC.



Example

A triangle has sides of length 10 cm and 11 cm and an area of 50 cm^2. Determine the two possible measures of the included angle. Give your answers accurate to 1 decimal place.

II- Cosine rule

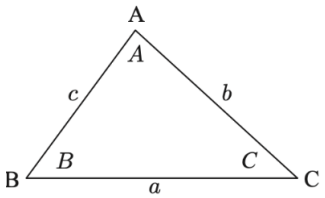
The cosine rule can be used to solve problems involving triangles given :

— two sides and an included angle

— three sides.

Property

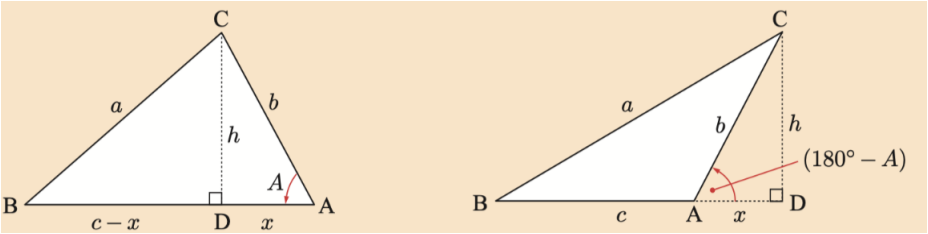
The cosine rule involves the sides and angles of a triangle.



In any $\triangle ABC$:

$a^2 = b^2 + c^2 - 2bc\cos A$ or $b^2 = a^2 + c^2 - 2ac\cos B$ or $c^2 = a^2 + b^2 - 2ab\cos C$.

Proof



In both triangles draw the altitude from C down to $[AB]$ (extended if necessary), meeting it at D. Let $AD = x$ and let $CD = h$.

Apply the theorem of Pythagoras in $\triangle BCD$:

$$a^2 = h^2 + (c - x)^2 \qquad \therefore \qquad a^2 = h^2 + c^2 - 2cx + x^2$$

$$a^2 = h^2 + (c + x)^2 \qquad \therefore \qquad a^2 = h^2 + c^2 + 2cx + x^2$$

In both cases, applying Pythagoras to $\triangle ADC$ gives

$$h^2 + x^2 = b^2.$$

$$\therefore \quad h^2 = b^2 - x^2,$$

and we substitute this into the equations above.

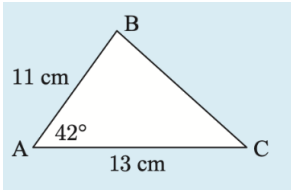
$$\therefore \quad a^2 = b^2 + c^2 - 2cx \qquad \text{or} \qquad a^2 = b^2 + c^2 + 2cx.$$

In $\triangle ADC$,

$$\cos A = \frac{x}{b} \quad \therefore \quad b\cos A = x \quad \therefore \quad a^2 = b^2 + c^2 - 2bc\cos A.$$

$$\cos(180^\circ - A) = \frac{x}{b} \quad \therefore \quad -\cos A = \frac{x}{b} \quad \therefore \quad -b\cos A = x \quad \therefore \quad a^2 = b^2 + c^2 - 2bc\cos A.$$

Example



Find, correct to 2 decimal places, the length of $[BC]$.

Example

In triangle ABC, $AB = 7$ cm, $BC = 5$ cm, and $CA = 8$ cm.

a. Find the measure of $\widehat{BCA}$.

b. Find the exact area of triangle ABC.

III- Sine rule

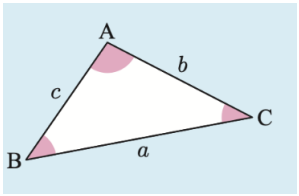
The sine rule is used to solve problems involving triangles, given :

— two angles and one side

— two sides and a non-included angle.

Property

In any triangle ABC with sides a, b, and c units in length, and opposite angles A, B, and C respectively,



$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} \quad \text{or} \quad \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

Proof

The area of any triangle ABC is given by

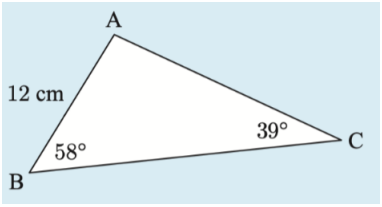
$$\frac{1}{2}bc\sin A = \frac{1}{2}ac\sin B = \frac{1}{2}ab\sin C.$$

Dividing each expression by $\frac{1}{2}abc$ gives

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}.$$

Example

Find the length of $[AC]$ correct to two decimal places.



Example

Find the measure of angle C in triangle ABC if $AC = 7$ cm, $AB = 11$ cm, and angle B measures 25°.

Example

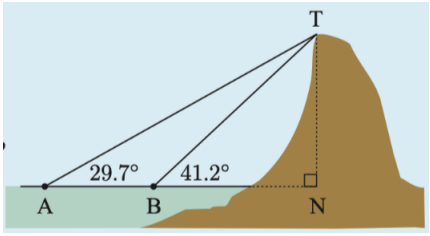
Find the measure of angle L in triangle KLM given that angle K measures 56°, $LM = 16.8$ m, and $KM = 13.5$ m.

IV- Applications

Example

The angles of elevation to the top of a mountain are measured from two beacons A and B at sea.

The measurements are shown on the diagram.



If the beacons are 1473 m apart, how high is the mountain ?

Example

Find the measure of $\widehat{RPV}$.

